

Good morning. I am Phichitchai Jueajan, with Pakkapol Phdungdan.

Our project is an automatic following and forming framework for inverter-based resources.

In fifteen minutes: the problem, three methods, the switching law, four results.

Point at: the section list.

Left: grid-following needs an external voltage-angle reference.

Right: grid-forming makes its own angle, and can replace the reference lost when the generator disconnects.

So neither mode is right all the time.

Point at: the last bullet on the right.

One in-house chain: power flow, small-signal stability, simulation. Four converters with selectable branches.

Third objective is the heart of it: decide the allowed grid-forming sets from the network condition, not by fixing the number in advance.

Point at: objective three.

Top row: power flow, then equilibrium, then screening, then integration — all on one operating point.

Bottom row back: check severity, then make the transaction.

The rule: commit only after current continuity and network convergence.

Point at: the arrow back to the transaction box.

IEEE 14-bus: one generator, four dual-mode converters.
 SG_1 at bus one; IBR_1 to IBR_4 at buses two, three, six and eight.

Every later figure is labelled by resource, not by bus.

Point at: the resource table.

Net complex power at a bus: the voltage times the conjugate of the network current. In polar form, the two real equations for P and Q .

The table gives what each bus type fixes.

Below: Newton–Raphson — mismatches, solve the boxed system, update, repeat.

Point at: the boxed Jacobian system.

We linearise the same equations the simulator uses.

Since $\mathbf{g}_y$ is invertible, a Schur complement gives the boxed reduced matrix $\mathbf{A}_{red}$.

Its eigenvalues give frequency and damping. Every physical root must have a negative real part.

Point at: the two boxed equations.

Implicit trapezoidal for the states, with the network enforced at the new time point — solved together.

The list on the right is what makes an event trustworthy: land exactly on it, apply the change atomically, require convergence after, roll back if it fails.

Point at: the event-handling block.

Each converter carries a fixed seventeen coordinates: shared plant, a grid-following branch, a grid-forming branch.

The inactive branch is held, not reset — that is why the mode can change mid-run.

Right: the PLL, and the virtual synchronous machine.

Point at: the two active-set expressions.

Severity S is the average of a voltage index and a frequency index, saturated to zero-to-one.

The command is deliberately asymmetric: into forming at 0.65 held 0.10 seconds — fast; back to following below 0.35 held a full second — slow.

And a gate: the current must match to 10^{-10} per unit.

Point at: the two dwell arrows.

Any single converter passes. Two pairs pass, and a pair gives the best margin. Every triple is rejected. All four together passes again.

Three fails and four passes — admissibility is non-monotonic in the count.

So every candidate must be computed, not inferred.

Point at: the three-converter row, then the four-converter row.

Six periods over 250 seconds: the generator trips at 20, load steps at 50, fault at 85, line trip at 110, reclose commanded at 145.

The two times under period six are results, read from the event log — not settings.

Point at: the two result times under period six.

Panels (a) and (b): the converters pick up the island after 20 seconds, then hand it back.

Panel (c): each disturbance drives severity above Γ_{on} , and between events it falls below Γ_{off} — so this is measurement, not a timetable.

Panel (d): the reference moves to IBR_1 , then back.

Point at: a peak in (c), then the matching step in (d).

Blue is adaptive, grey is fixed following. Bus nine carries load but no device, so it shows what the island delivers.

In (a) to (c) blue holds near one per unit; grey sags to about two-thirds and delivers less.

In (d), the honest trade-off: grey holds a narrower frequency band.

Point at: the gap in (a), then the band in (d).

Three findings: after the generator disconnects at least one forming converter is required; admissibility is non-monotonic; staged switching reaches 250 seconds and hands everything back.

The box compares the arms — locked following stops at 20 seconds, four pinned units fail at 25.485.

Thank you — we welcome questions.

Point at: the comparative-outcome box.